① A parallel beam of light of wavelength 6000 Å is incident on a plain transparent film of R.I 1.5. If angle of refraction is 28°. Find the min thickness of the film if it appears bright in the reflected light. Take n=1

Solⁿ:-

$\lambda = 6000\,Å = 6\times10^{-7}\,m$, $\mu = 1.5$, $r = 28^\circ$, $t = ?$, $n = 1$

for bright in the reflected system

$$2\mu t \cos r = (2n-1)\frac{\lambda}{2} \qquad n = 1, 2, 3, \dots$$

for $t = t_{min}$, $n = 1$

$$2\mu t_{min} \cos r = \frac{\lambda}{2}$$

$$\therefore \quad t_{min} = \frac{\lambda}{4\mu \cos r} = \frac{6\times10^{-7}}{4\times1.5\times\cos 28^\circ}$$

$$t_{min} = \frac{6\times10^{-7}}{4\times1.5\times0.8829}$$

$$t_{min} = \frac{6\times10^{-7}}{5.2976} = 1.132\times10^{-7}\,m$$

$$t_{min} = \underline{\underline{1132\,Å}}$$

4'

G'

② A soap film of RI 1.33 and thickness 1.5×10^{-5} cm is illuminated by light incident at 30°. Light reflected from it shows a dark band in 2^{nd} order. calculate wavelength corresponding to the dark band.

<u>Solⁿ</u>: $\mu = 1.33$, $t = 1.5 \times 10^{-5}$ cm $= 1.5 \times 10^{-7}$ m

$i = 30^\circ$, dark, $n = 2$, $\lambda = ?$

for dark in reflected system

$$2\mu t \cos r = n\lambda \longrightarrow ①$$

$$\mu = \frac{\sin i}{\sin r} \Rightarrow \sin r = \frac{\sin i}{\mu}$$

$$\sin r = \frac{\sin 30^\circ}{1.33} = \frac{0.5}{1.33} = 0.3759$$

$$\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - (0.3759)^2}$$

$$\cos r = \sqrt{1 - 0.1413} = 0.9266$$

from ①

$$\lambda = \frac{2\mu t \cos r}{n} = \frac{2 \times 1.33 \times 1.5 \times 10^{-7} \times 0.9266}{2}$$

$$\lambda = 1.848 \times 10^{-7} \text{ m}$$

$$\lambda = 1848 \text{ Å}$$

(3) A parallel beam of light falls normally on an oil film of RI 1.25. Complete destructive interference is observed for wavelengths 5000Å and 6000Å and for no wavelength in between. Find the thickness of the oil.

Soln:. $\mu = 1.25$, $\lambda_1 = 5000\,\text{Å} = 5\times10^{-7}\,m$, $r = 0$

$\lambda_2 = 6000\,\text{Å} = 6\times10^{-7}\,m$

$t = ?$

for destructive interference in reflected system

$2\mu t \cos r = n\lambda$

$\because r = 0$, $\cos r = 1$

$2\mu t = n\lambda$

for λ_1 and λ_2 $\quad \because \lambda_1 < \lambda_2$

$\lambda_1 \downarrow (n+1)$, $\lambda_2 \downarrow (n)$

$\therefore 2\mu t = (n+1)\lambda_1 \longrightarrow (1)$

and $2\mu t = n\lambda_2 \longrightarrow (2)$

$\therefore$ from (1) and (2)

$(n+1)\lambda_1 = n\lambda_2$

$n\lambda_1 + \lambda_1 = n\lambda_2$

$n(\lambda_2 - \lambda_1) = \lambda_1$

$n = \dfrac{\lambda_1}{(\lambda_2 - \lambda_1)} = \dfrac{5\times10^{-7}}{1\times10^{-7}}$

$\therefore \boxed{n = 5}$

$\therefore$ from (1) $\quad t = \dfrac{6\times5\times10^{-7}}{2\times1.25} = 12\times10^{-7}\,m$

$t = 12000\,\text{Å}$

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(4) White light falls normally on a soap film ($\mu = 1.33$) of thickness 3800 Å. which wavelength/s within the visible spectrum (4000 - 7000 Å) will be intensified in the reflected light?

Sol^n: $\mu = 1.33$, $t = 3800$ Å $= 3.8 \times 10^{-7}$ m

$i = 0$, $r = 0$

for max. in the reflected system

$$2\mu t \cos r = (2n-1)\frac{\lambda}{2} \qquad n = 1, 2, 3, \dots$$

$\because$ $r = 0$

$$2\mu t = (2n-1)\frac{\lambda}{2}$$

$$\therefore \lambda = \frac{4\mu t}{(2n-1)} = \frac{4 \times 1.33 \times 3.8 \times 10^{-7}}{(2n-1)}$$

$$\lambda = \frac{20.216 \times 10^{-7}}{(2n-1)}$$

when $n = 1$, $\lambda = 20.216 \times 10^{-7}$ m

$\lambda = 20216$ Å (not in visible)

when $n = 2$, $\lambda = 6.738 \times 10^{-7}$ m

$\lambda = 6738$ Å (not in visible)

when $n = 3$, $\lambda = 4.043 \times 10^{-7}$ m

$\lambda = 4043$ Å (visible)

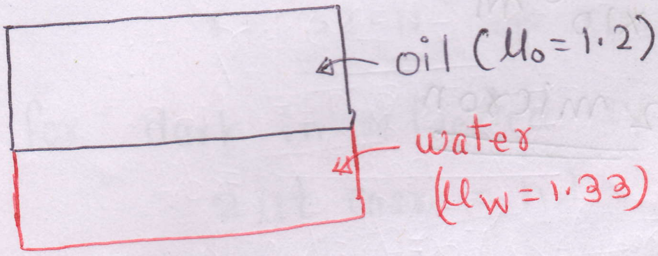
when $n = 4$, $\lambda = 2.888 \times 10^{-7}$ m

$\lambda = 2888$ Å (not in visible)

⑤ A parallel beam of light falls normally on an oil film of R.I 1.2 having uniform thickness which is spread on water (RI 1.33). Brightness is obtained for wavelengths 5000 Å and 7500 Å and for no wavelength in between. Find thickness of the oil film.

$\mu_o = 1.2, \ \mu_w = 1.33, \ \lambda_1 = 5000\ \text{Å} = 5\times10^{-7}\ m$

$\lambda_2 = 7500\ \text{Å} = 7.5\times10^{-7}\ m$



$\lambda_1 < \lambda_2$

$\downarrow \qquad \downarrow$

$(n+1) \qquad (n)$

$\delta = 2\mu t \cos r$

for max / bright $\delta = n\lambda$

$\therefore 2\mu t \cos r = n\lambda$

$\therefore 2\mu t = n\lambda$

$\therefore$ for λ_1 and λ_2

$2\mu_o t_o = n\lambda_2 \longrightarrow ①$

$2\mu_o t_o = (n+1)\lambda_1 \longrightarrow ②$

$\therefore$ from ① and ②, $n\lambda_2 = (n+1)\lambda_1$

$n\lambda_2 = n\lambda_1 + \lambda_1$

$\Rightarrow n = \dfrac{\lambda_1}{(\lambda_2 - \lambda_1)} = \dfrac{5\times10^{-7}}{2.5\times10^{-7}} = 2$

$\boxed{n=2}$

from ① $t_o = \dfrac{2\times7.5\times10^{-7}}{2\times1.2} = 6.25\times10^{-7}\ m.$

$t_o = \underline{\underline{6250\ \text{Å}}}$

⑥ A soap of RI 1.33 is illuminated by with light of different wavelengths at an angle of 45°. calculate the smallest thickness of the film which will appear dark by reflection. Wavelength of light used is 5890 Å.

Sol^n: $\mu = 1.33$, $i = 45°$, $t_{min} = ?$, $\lambda = 5.890 \times 10^{-7}$ m

$$\mu = \frac{\sin i}{\sin r} \Rightarrow \sin r = \frac{\sin i}{\mu} = \frac{\sin 45°}{1.33}$$

$$\sin r = \frac{0.707}{1.33} = 0.5316$$

$$r = \underline{\underline{32.11°}} \Rightarrow \cos r = 0.8469$$

for dark in reflected system

$$2\mu t \cos r = n\lambda$$

$$t = \frac{n\lambda}{2\mu \cos r}$$

$$t_{min} = \frac{1 \times 5.890 \times 10^{-7}}{2 \times 1.33 \times \cos 32.11°}$$

$$= \frac{5.890 \times 10^{-7}}{2.253} = 2.6142 \times 10^{-7}$$

$$t_{min} = \underline{\underline{2614.2 \text{ Å}}}$$

(7) 6 A soap film of RI 1.33 and thickness 1.50 μm is illuminated by white light incident at an angle of 45^0. In the reflected light a dark band is observed for the wavelength 5×10^{-5} cm. calculate the order of the interference band.

sol^n: $\mu = 1.33$, $t = 150\ \mu m = 1.50 \times 10^{-6}\ m$

$i = 45^0$, $\lambda = 5 \times 10^{-7}\ m$, $n = ?$

$\mu = \frac{\sin i}{\sin r} \Rightarrow \sin r = \frac{\sin i}{\mu} = \frac{\sin 45^0}{1.33}$

$\sin r = 0.5316$

$r = 32.11^0$

$\cos r = 0.8469$

for dark in reflected system

$2\mu t \cos r = n\lambda$

$n = \frac{2\mu t \cos r}{\lambda} = \frac{2 \times 1.33 \times 1.5 \times 10^{-6} \times 0.8469}{5 \times 10^{-7}}$

$n = 6.75$

$\therefore$ highest possible order is 6

(3)

8. A drop of liquid of volume 0.2cc is spread over the whole surface of tank of water of area 1 sq.m forming a thin film. When white light is incident normally on the film a dark band corresponding to the wavelength 5500 Å is seen the spectrum. Find the refractive index of liquid.

Consider $\mu_e > \mu_w$

Sol^n:

Volume $= 0.2\ cc = 0.2 \times 10^{-3}\ m^3$

area $= 1\ m^2$

$\therefore$ thickness of film $= 0.2 \times 10^{-6}\ m^2$

dark, $\lambda = 5.5 \times 10^{-7}\ m$

for dark in reflected system

$2\mu t \cos r = n\lambda$, $r = 0$, $\cos r = 1$

$$\mu = \frac{n\lambda}{2t\cos r} \quad \text{(let } n=1\text{)}$$

2

$$\mu = \frac{1 \times 5.5 \times 10^{-7}}{2 \times 0.2 \times 10^{-6} \times 1}$$

6

$$\mu = 1.375$$

⑨ A parallel beam of sodium light ($\lambda = 5890 \mathring{A}$) strikes a film of oil floating on water. When viewed at an angle of 60° from the surface 8th dark band is seen. If the RI of oil is 1.46, find the thickness of the film. (normally 1.5)

Sol^n :

$\mu_o > \mu_w$

$\lambda = 5.890 \times 10^{-7}$ m,

$i = 30^\circ$,

$n = 8$, $\mu = 1.46$

$t = ?$

for dark in reflected system

$$2\mu t \cos r = n\lambda$$

$$t = \frac{n\lambda}{2\mu \cos r} \longrightarrow ①$$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 30^\circ}{\sin r}$$

$$\therefore \sin r = \frac{1/2}{1.46} = 0.3424$$

$$r = 20.02^\circ$$

$$\cos r = \underline{\underline{0.9395}}$$

from ①

$$t = \frac{8 \times 5.890 \times 10^{-7}}{2 \times 1.46 \times 0.9395}$$

$$t = 1.7176 \times 10^{-6} \text{ m}$$

$$t = 17176 \times 10^{-10} \text{ m}$$

$$t = \underline{\underline{17176}} \mathring{A}$$

(10) Calculate the wavelength which would cut-off from reflection due to a film of thickness 1 micron and refractive index 1.28.

Sol^n:- $t = 1 \times 10^{-6}$ m, $\mu_f = 1.28$, $\lambda = ?$

$$t = \frac{\lambda}{4\mu_f}$$

$$\Rightarrow \lambda = (4)(\mu_f)(t)$$

$$\lambda = 4 \times 1.28 \times 10^{-6}$$

$$\lambda = 5.12 \times 10^{-6} \text{ m}$$

$$\lambda = 5.12 \text{ micron}$$

(11) Can a thin film of MgF_2 of R.I 1.22 act as an antireflection film if deposited on glass of R.I 1.52? If yes, determine the minimum thickness required to cut-off reflection due to wavelength 5500 Å

Soln:

$\lambda = 5500\ \text{Å} = 5.5 \times 10^{-7}$ m,

$\mu_f = \sqrt{\mu_g} = \sqrt{1.52} = 1.23$

$\mu_f \approx \sqrt{\mu_g}$

$\Rightarrow$ Yes, MgF_2 can act as an antireflection film.

$$t = \frac{\lambda}{4\mu_f} = \frac{5.5 \times 10^{-7}}{4 \times 1.22}$$

(1)

$$t = 1.12 \times 10^{-7}\ \text{m}$$

$$t = \underline{\underline{1120\ \text{Å}}}$$

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